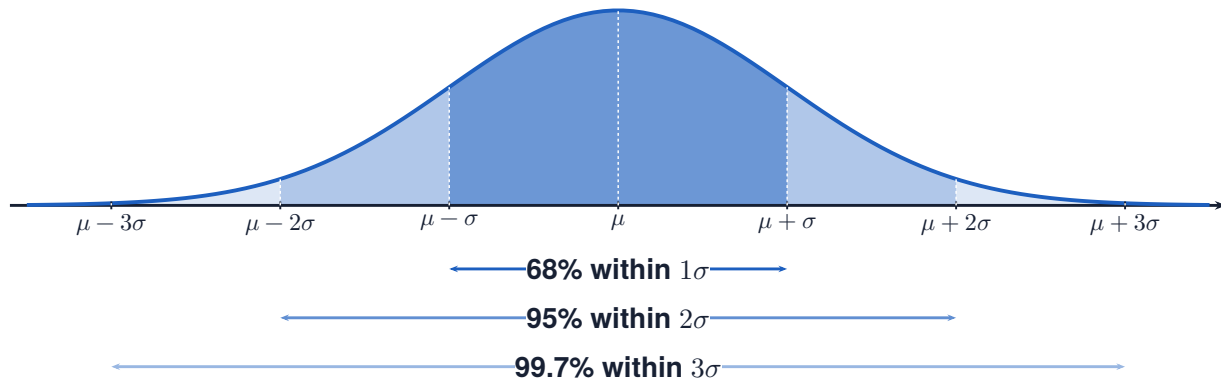


THE NORMAL DISTRIBUTION

new 2.11 • old 1.10

ZONE A · THE CURVE + THE FORMULA



$$z = \frac{x - \mu}{\sigma}$$

 $x \rightarrow z \rightarrow \text{area}$

normalcdf: value to probability

 $\text{area} \rightarrow z \rightarrow x$

invNorm: probability to value

ZONE B · WHAT THE SYMBOLS MEAN

CENTER + SPREAD

$$X \sim N(\mu, \sigma)$$

 μ : mean σ : standard deviation

Total area under the curve is 1.

STANDARDIZE THE DISTANCE

 z counts SDs from the mean.Positive: above μ . Negative: below μ .At the mean, $z = 0$.

ZONE C · SAY IT IN WORDS

"_____ is _____ standard deviations [above / below] the mean of _____."

"About _____% of _____ have _____ below _____."

ZONE D · WATCH OUT

DRAW BEFORE YOU CALCULATE

Draw the curve. Mark the value.
Shade the requested region first.

PERCENTILE = AREA TO THE LEFT

Multiply the left-tail area by 100.

The 68–95–99.7 rule gives approximate percentages for a normal model.